

(56)

References Cited

U.S. PATENT DOCUMENTS

6,647,346	B1	11/2003	Zikes	
7,100,382	B2	9/2006	Butler et al.	
7,444,824	B1	11/2008	Butler et al.	
7,464,561	B1	12/2008	Butler et al.	
7,694,525	B2	4/2010	Butler et al.	
8,699,973	B2 *	4/2014	Southcombe	 H03F 3/72 455/127.2
9,484,771	B2	11/2016	Braylovskiy et al.	
9,837,816	B2	12/2017	Miyamura et al.	
10,605,495	B1	3/2020	Falls et al.	
10,840,735	B1	11/2020	Cooper	
2013/0079959	A1 *	3/2013	Swanson	 B61C 17/12 701/19
2013/0107401	A1	5/2013	Helt	
2013/0211615	A1	8/2013	Zikes et al.	
2013/0249429	A1 *	9/2013	Woytowicz	 H05B 47/165 315/246
2015/0060038	A1 *	3/2015	Lau	 H02P 7/295 165/267
2018/0292102	A1	10/2018	Mowris et al.	
2020/0194205	A1	6/2020	Buescher et al.	

OTHER PUBLICATIONS

49P11-843 Emerson SureSwitch™ Installation Instructions; web accessed Dec. 14, 2018; 4 pgs.
 SureSwitch™ Multi-Voltage Contactor Upgrade; web accessed Dec. 14, 2018; 2 pgs.

* cited by examiner

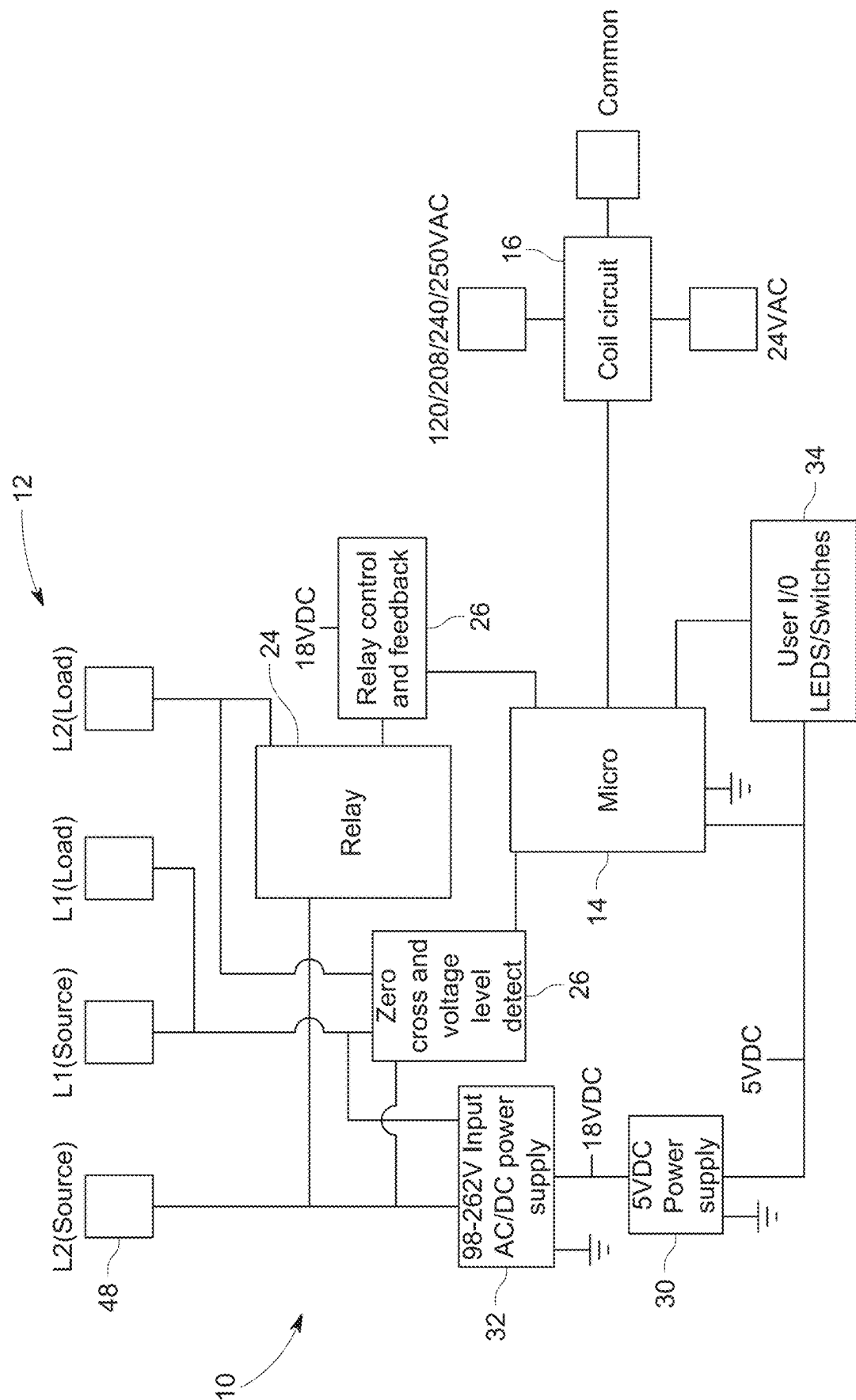


FIG. 1

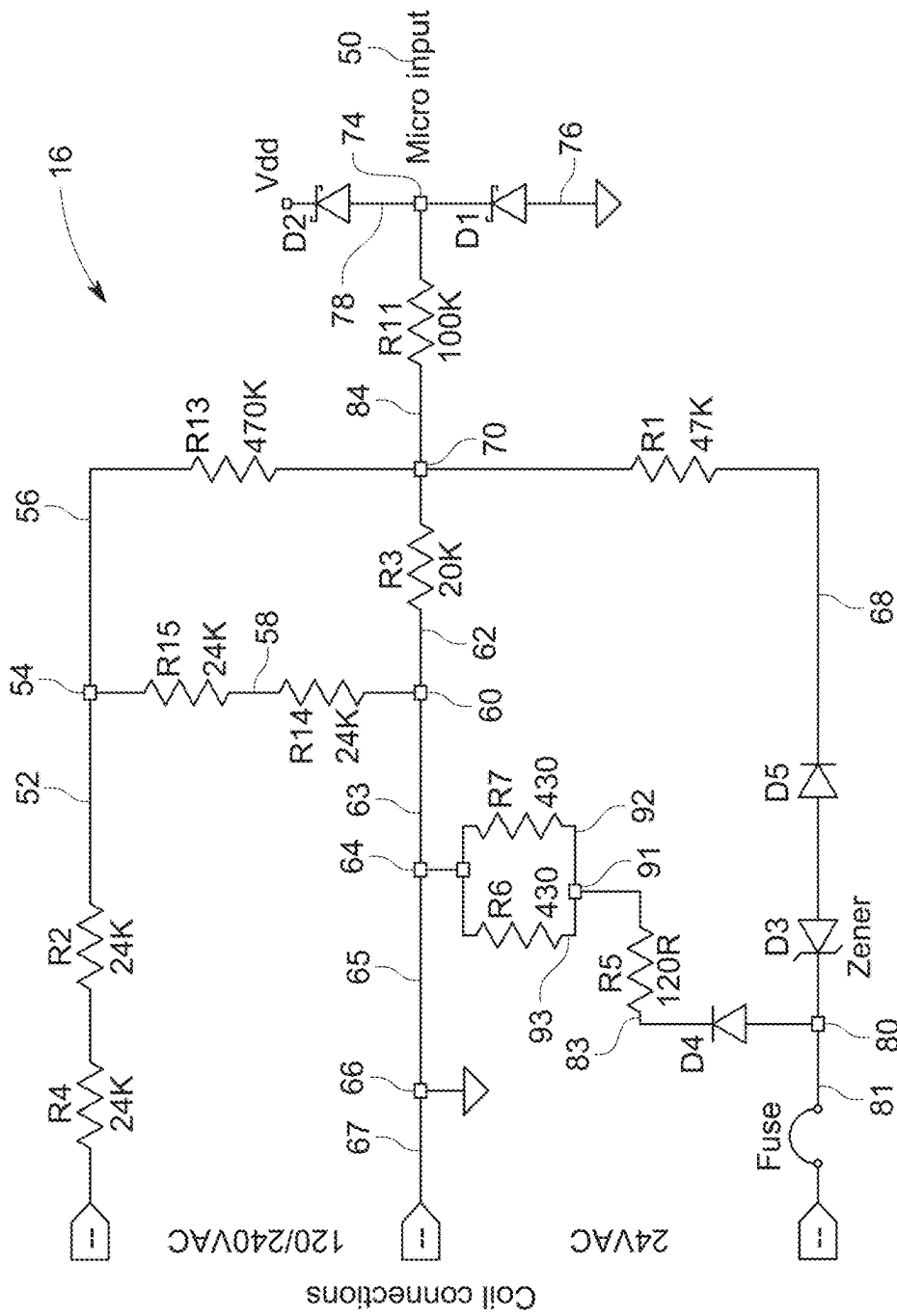


FIG. 2

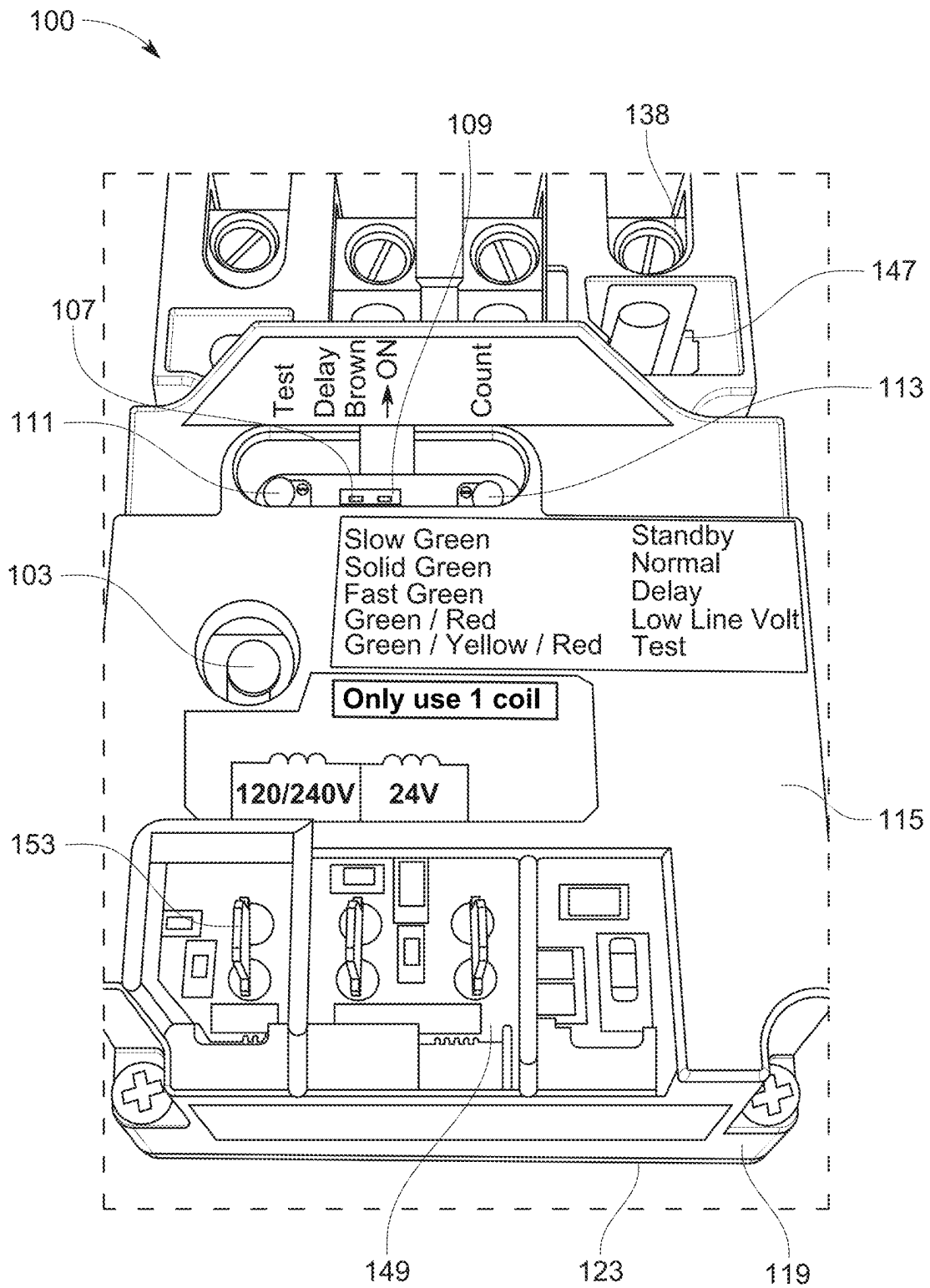
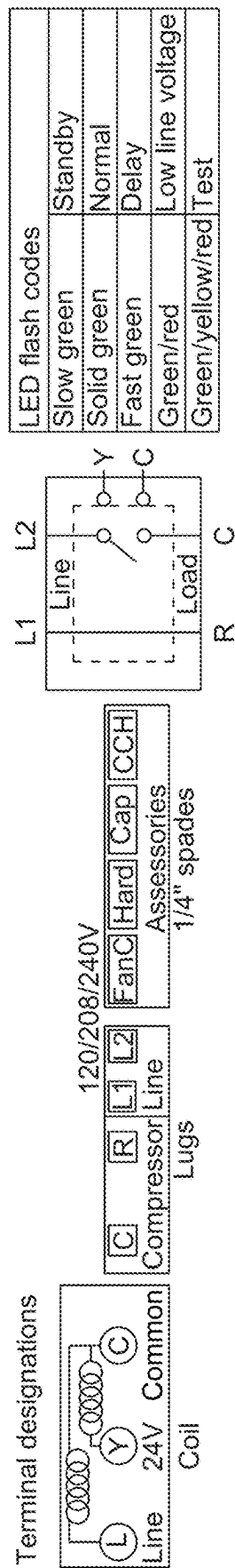


FIG. 3



Coil Details: For voltage over 100 VAC: Connect line level control voltage to "Line", connect neutral or opposite leg to "Common"
For class 2 24VAC: Connect 24V control voltage to "Y", connect 24V ground to "Common"

FIG. 4

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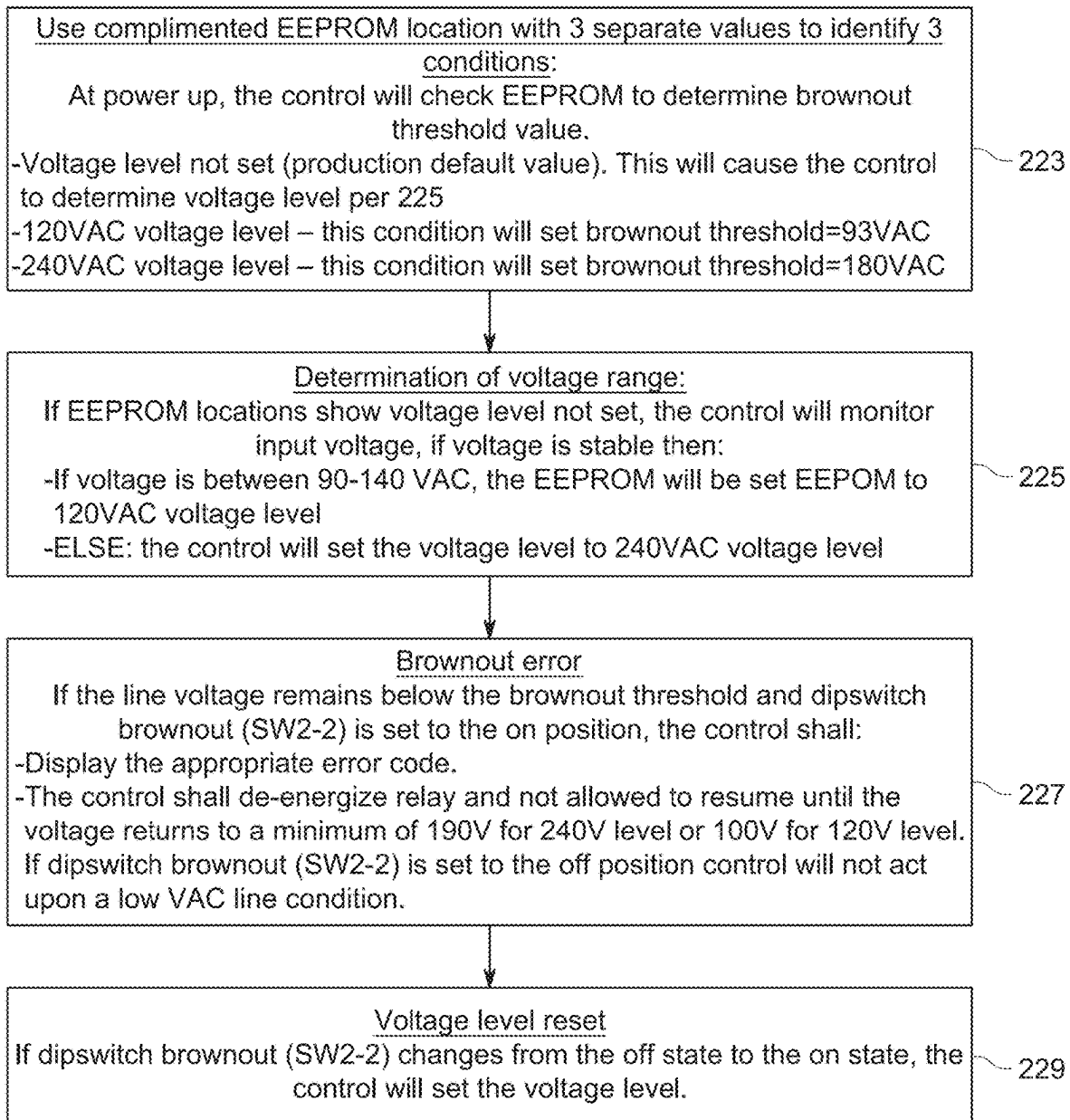


FIG. 5

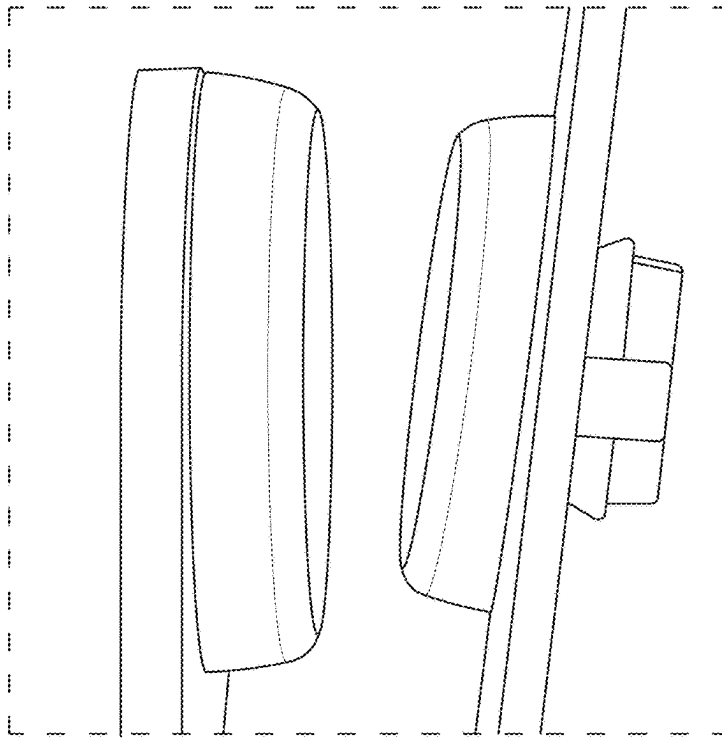


FIG. 6

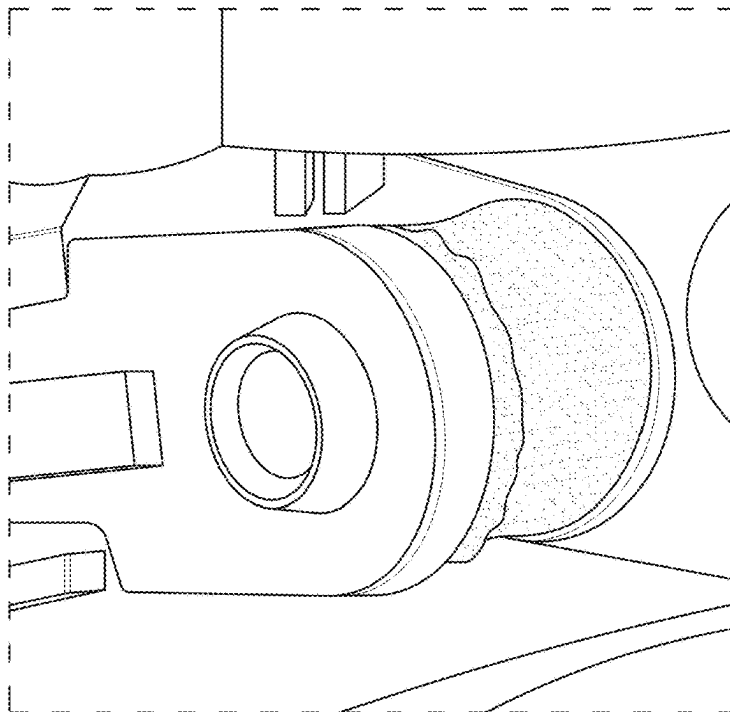


FIG. 7
(PRIOR ART)

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MULTI-VOLTAGE CONTACTORS, CONTROLS, AND RELATED METHODS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of allowed U.S. Non-Provisional application Ser. No. 16/691,095 filed Nov. 21, 2019, which published as US2020/0194205 on Jun. 18, 2020.

U.S. Non-Provisional application Ser. No. 16/691,095 claims the benefit and priority of U.S. Provisional Application No. 62/781,461 filed Dec. 18, 2018.

The above applications are incorporated herein by reference in their entirety.

FIELD

The present disclosure relates to multi-voltage contactors, controls, and related methods.

BACKGROUND

This section provides background information related to the present disclosure which is not necessarily prior art.

Climate control systems (e.g., an air conditioning, heat pump systems, refrigeration systems, etc.) typically include components such as compressors that are turned on and off by contactors in response to thermostat signals. Such contactors are relatively expensive, and frequently provide no functionality except to connect and disconnect system components to and from electric power. Also, contactors are common repair items due to the open nature of the electrical contact and the high current inductive loads that are switched, which can burn contacts.

DRAWINGS

The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations, and are not intended to limit the scope of the present disclosure.

FIG. 1 is a schematic wiring diagram illustrating an exemplary embodiment of a relay switch control connected with components of an HVAC system in accordance with an exemplary implementation of the present disclosure.

FIG. 2 is a schematic wiring diagram illustrating an exemplary embodiment of a “coil” control circuit in accordance with an exemplary implementation of the present disclosure.

FIG. 3 shows an exemplary embodiment of a relay switch control, which may include the circuits shown in FIGS. 1 and 2 in accordance with an exemplary implementation of the present disclosure.

FIG. 4 illustrates exemplary terminal designations and LED flash codes that may be used with the relay switch control shown in FIG. 3 in accordance with an exemplary implementation of the present disclosure.

FIG. 5 is a flow chart illustrating an example Brownout Auto Detect method for determining input voltage and adjusting brownout threshold according to an example embodiment of the present disclosure.

FIG. 6 shows an exemplary embodiment of a contactor after one million cycles using microprocessor controlled switching algorithms that reduce arcing and virtually eliminates contact pitting and welding.

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FIG. 7 shows a traditional contactor shown after 300,000 cycles, whereby a comparison of FIGS. 5 and 6 shows a performance improvement that may be realized by using microprocessor controlled switching algorithms in accordance with exemplary implementations of the present disclosure.

Corresponding reference numerals may indicate corresponding (though not necessarily identical) parts throughout the several views of the drawings.

DETAILED DESCRIPTION

Example embodiments will now be described more fully with reference to the accompanying drawings.

Exemplary embodiments are disclosed that include multi-voltage contactors. The multi-voltage or universal contactors may be configured to be operable across or with a range of activation inputs, such as activation inputs ranging from 98 VAC to 276 VAC inputs (e.g., 120, 208, 240, 250, 24 VAC inputs, etc.), etc. For example, a multi-voltage contactor disclosed herein may be configured to accept a 120, 208, 240, 250, or 24 VAC activation input to switch loads of the same or different voltages. By way of comparison, an existing residential cooling specific design of a printed circuit board (PCB) mounted relay capable of high current compressor switching may be configured to only accept a 24 VAC input.

In exemplary embodiments, a relay switch control may include a circuit similar or identical to a circuit of a microprocessor-controlled replacement for a standard contactor as disclosed in U.S. Patent Application US2013/0211615, the entire disclosure of which is incorporated herein by reference. In exemplary embodiments disclosed herein, the relay switch control may be configured to include the following features:

The power supply may be configured to run on 98-276 VAC inputs.

The coil/control circuit may be configured to run on 120/208/240/250V input or 24V input, with ground or neutral reference.

A firmware algorithm may be provided to account for brownout.

A firmware algorithm may be provided to account for phase difference between the switched voltage and the coil voltage. This may include a routine that samples throughout the line cycle and looks for a balance of high and low signals indicating an AC signal.

A common connector and two potential power connections may be provided for the “coil” connection. A sliding plastic door may help prevent miswiring such that only one of the power connections (120/208/240/250V or 24V) is available to connect at a time. The door may only enable connection to the low voltage AC power or the line voltage source.

A wiring box may be integrated into a plastic enclosure with the potential for an optional compressor switch.

A method of detecting AC voltage may also be included.

With the ability to accept a range of activation inputs (e.g., 98-276 VAC inputs, or 120, 208, 240, 250, or 24 VAC inputs, etc.) to switch loads of the same or different voltages, exemplary embodiments of the multi-voltage contactors disclosed herein may be used to replace multiple different voltage-specific contactors. For example, the multi-voltage contactors disclosed herein may be used as a multi-voltage electronic replacement for mechanical compressor contactors, which typically are voltage specific on the coil side.

Exemplary embodiments may also provide benefits of an enclosed PCB mounted relay with zero cross capability that can be used on multiple voltages and phases. Exemplary embodiments may also include an integrated wiring box that allows for reduced number of parts required by the original equipment manufacturer (OEM).

FIG. 1 illustrates an exemplary embodiment of a relay switch control **10** embodying one or more aspects of the present disclosure. The relay switch control **10** is connected with components of an HVAC system indicated generally by reference number **12**. The control **10** includes a microprocessor **14**, which may include firmware changes for brown-out protection and/or for managing phase mismatch of a control circuit. The microprocessor **14** is configured to receive control signals (e.g., signals from an indoor thermostat (not shown), etc.) from a coil circuit **16**. For example, the microprocessor **14** may be coupled for communication with and receive control signals via the micro input/out (**IO**) **50** of the coil control circuit **16** (FIG. 2).

The control **10** includes a relay **24** electrically connected with line voltage source **L2** and load **L2** (e.g., compressor motor, etc.). The relay **24** is also electrically connected with relay control and feedback **26**. In turn, the relay control and feedback **26** is electrically connected with the microprocessor **14**. The relay **24** is operable by the microprocessor **14** via the relay control and feedback **26** to electrically connect or disconnect the line voltage source **L2** and load **L2**. The relay **24** may be substantially enclosed in a seal (e.g., a coating of epoxy glue, etc.) that is configured to prevent the intrusion of foreign objects (e.g., insects, debris, contaminants, etc.) into contacts (not shown) of the relay **24**.

In the illustrated embodiment, the relay **24** is a latching relay. The control **10** may include zero cross and voltage level detect **26**. With the aid, e.g., of optical sensing of sparking at the relay **24**, the microprocessor **14** may be configured to provide “zero cross” switching of current through the relay **24** such that current is switched through the relay **24** at or very close to zero crossing of the line voltage. Such switching may be performed as disclosed in U.S. Pat. No. 7,464,561, the entire disclosure of which is incorporated herein by reference. Arcing and contact damage to the relay **24** may thereby be reduced or eliminated.

Also shown in FIG. 1 are a 5 VDC power supply **30**, a 98-276V input AC/DC power supply **32**, and user input/out devices **34**. The user input/out devices **34** may include switches (e.g., dipswitches, push button switches, other switches, etc.) and LEDs (e.g., multi-colored LEDs, other light sources, etc.). For example, the control **10** may include two dipswitches disposed between two push buttons. Fewer or additional dipswitches and/or push buttons may be provided in other exemplary embodiments.

The PCB and housing of the relay switch control may be configured to accommodate for the potential line voltage connections, e.g., 24 VAC, 120/208/240/250 VAC, etc. In exemplary embodiments, the control **10** may be configured to be operable across or with a range of activation inputs, such as activation inputs ranging from 98 VAC to 276 VAC inputs (e.g., 120, 208, 240, 250, 24 VAC inputs, etc.) to switch loads of the same or different voltages.

A crankcase heater may also be connected to line voltages. The crankcase heater may be, e.g., a “belly band” crankcase heater. The control **10** may also be connected with a fan motor, a fan capacitor, and a compressor capacitor. R and C terminals of the control **10** may be connected, e.g., via lug type connectors **48**, with R (run) and C (common) terminals of a compressor motor, etc. An S (start) terminal of the compressor motor may be connected with a HERM

terminal of the compressor capacitor. The control may be connected with a C (common) terminal of the fan motor. In this example, the control **10** may switch the fan motor on or off with the compressor motor through the relay **24**. The fan motor may be, but is not limited to, e.g., a one-speed permanent split capacitor (PSC) motor for an outdoor fan, etc. R (run) and S (start) terminals of the fan motor may be connected with the fan capacitor. The example control **10** may be configured to be compatible with most, if not all, types of single-speed PSC outdoor fan motor wiring including 3-wire, 4-wire, and universal replacement motors.

The example control **10** may also be configured so that it is compatible with both dual capacitor (separate compressor and outdoor fan) systems and single capacitor (combined compressor and outdoor fan) systems. The example control **10** may be further configured to be compatible with both 2-wire and 3-wire hard start kits.

FIG. 2 illustrates an exemplary embodiment of a “coil” control circuit **16** embodying one or more aspects of the present disclosure. The “coil” control circuit **16** includes a micro input/output **50**, which may be coupled with and transmit control signals to the microprocessor **14** of the relay switch control **10** (FIG. 1).

As shown in FIG. 2 the “coil” control circuit **16** includes terminals or coil connections labeled 120/240 VAC and 24 VAC. For a 24 VAC coil, the coil wires are connected to the 24 VAC terminal and a common terminal. For a line voltage coil, the coil wires are connected to the 120/240 VAC terminal and the common terminal. Accordingly, this illustrated exemplary embodiment is configured to accept the 120/240 VAC or 24 VAC activation input and switch loads of the same or different voltages. But as explained herein, exemplary embodiments may be configured to be operable across or with a range of activation inputs, such as activation inputs ranging from 98 VAC to 276 VAC inputs (e.g., 120, 208, 240, 250, 24 VAC inputs, etc.) to switch loads of the same or different voltages.

The “coil” control circuit **16** includes a fuse or protective device **17** in the current path of the 24 VAC control signal. The fuse or protective device may protect against miswiring.

The “coil” control circuit **16** further includes a plurality of diodes and a plurality of resistors. In this exemplary embodiment, the “coil” control circuit **16** includes five diodes **D1**, **D2**, **D3**, **D4**, and **D5**. By way of example only, diodes **D1** and **D2** may comprise Schottky diodes (e.g., 5 VDC Schottky diodes, etc.). Diode **D3** may comprise a Zener diode that regulates current flow through net **68**. Diodes **D4** and **D5** may comprise standard diodes to block current flow in one direction. The diodes **D3** and **D4** are electrically connected with the fuse via the junction **80** of the circuit paths **81**, **83**, and **68**. Alternative embodiments may include more or less than five diodes and/or different diode configurations.

Also in this exemplary embodiment, the “coil” control circuit **16** include eleven resistors **R1**, **R2**, **R3**, **R4**, **R5**, **R6**, **R7**, **R11**, **R13**, **R14**, and **R15**. By way of example, resistor **R1** may comprise a 47 k ohm resistor. Resistors **R2**, **R4**, and **R15** may each comprise a 24 k ohm resistor. Resistor **R13** may comprise a 470 k ohm resistor. Resistor **R14** may comprise a 24 k ohm resistor. Resistors **R6** and **R7** may each comprise a 430 ohm resistor. Resistor **R5** may comprise a 120 ohm resistor. Resistor **R3** may comprise a 20 K ohm resistor. Resistor **R11** may comprise a 47 k ohm resistor. The resistance values for the eleven resistors are provided for purpose of illustration only. Alternative embodiments may include more or less than fifteen resistors and/or different resistor configurations.

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With continued reference to FIG. 2, the 120/240 VAC terminal is electrically connected with resistors R4 along a circuit path 52. Resistor R4 is also electrically connected in series with resistor R2 along the circuit path 52.

Resistor R2 is electrically connected with resistor R15 and with the resistor R13 via a junction 54 of the circuit paths 52, 56, 58 that respectively include resistors R2, R13, and R15.

Resistor R15 is electrically connected in series with R14 along the circuit path 58. Resistor 14 is electrically connected with resistor R3 via junction 60 and circuit path 62. Resistor 14 is also electrically connected with the 120/240 VAC terminal via junctions 64, 66 and circuit paths 63, 65, 67.

The 120/240 VAC terminal is electrically connected with ground via the junction 66 and circuit path 67. The 120/240 VAC terminal is also electrically connected with the fuse electrically connected to the 24 VAC terminal via the junctions 66, 64, 91, 80, circuit paths 67, 65, 92, 93, 83, and 81, resistors R6, R7, R5, and diode D4 as shown in FIG. 2.

Resistor R13 is electrically connected with resistors R1, R3, and R11 via a junction 70 of the circuit paths 56, 62, 68, and 84 that respectively include resistors R13, R3, R1, and R11.

Resistor R11 is electrically connected with diode D1, diode D2, and the micro input/output 50 via a junction 74 of the circuit paths 84, 76, 78 that respectively include resistor R11, diode D1, and diode D2.

Diode D5 is electrically connected in series with diode D3 along the circuit path 68.

Diode D3 is electrically connected with diode D4 and with the fuse via the junction 80 of the circuit paths 68, 81, and 83 that respectively include diode D3, the fuse, and the diode D4.

Diode D4 is electrically connected in series with resistor R5 along circuit path 83. In turn, resistor R5 is electrically connected with resistor R6 and with resistor R7 via a junction 91 of the circuit paths 83, 92, and 93 that respectively include resistor R5, resistor R6, and resistor R7.

FIG. 3 shows an exemplary embodiment of a relay switch control 100 embodying one or more aspects of the present disclosure. The control 100 may include a microprocessor and sealed relay, e.g., circuits 10 and 16 as described with reference to FIGS. 1 and 2, respectively.

As shown in FIG. 3, the control 100 includes an indicator light 103 (e.g., a multi-color LED, other light source, etc.), which in the present example embodiment is a tri-color LED. The indicator light 103 is operable by a microprocessor (not shown) of the control 100 to indicate faults, status, and/or the number of cycles through which the relay has cycled.

In this exemplary embodiment, the control 100 includes two dipswitches 107 and 109. By way of example, the first dipswitch 107 may be used to select/set a short cycle delay, and the second dipswitch 109 may be used to select or deselect brownout protection. Also in this exemplary embodiment, the control 100 includes two push buttons 111 and 113 respectively indicated as "TEST" and "COUNT". An example operation and functionality of the onboard push buttons 111 and 113 ("TEST" and "COUNT") is described below.

The control 100 includes a two-piece housing 115, e.g., a two-piece plastic housing with integral mounting features, etc. The two-piece housing 115 includes an upper housing portion or cover 119 and a lower housing portion 123. A microprocessor and sealed relay are provided beneath the upper cover 119, and therefore are not visible in FIG. 3. The

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housing 115 may include openings in the upper housing portion or cover 119 for terminal connections and connections to a compressor and fan, etc. Lug connectors 138 are provided for line voltage inputs and outputs, e.g., as discussed above with reference to FIG. 1, etc. Connectors 147 are provided for connection of compressor and fan capacitors, fan, etc. to line voltages (L1, L2) e.g., as discussed above with reference to FIG. 1, etc.

The control 100 includes a printed circuit board (PCB) 149 on which the microprocessor and sealed relay are provided. Although the PCB 149 is horizontally situated relative to the housing bottom portion 123, a PCB could be oriented in other directions, e.g., vertically within the housing in other relay switch control embodiments, etc. Connectors 153 are provided on the PCB 149 for connection of the control 100, e.g., with a thermostat (not shown).

In various embodiments, other or additional components of a climate control system may be connected with the control 100, including but not limited to high pressure and low pressure switches, temperature sensors for suction and/or liquid lines from a compressor, etc. In some example embodiments, an ambient temperature sensor may be connected with the control 100 to enable a comparison between ambient temperature and a threshold temperature to determine whether to switch off a heat pump compressor.

The control 100 may include mounting holes (e.g., four-hole mounting, etc.) configured to match with existing standard contactor holes for mounting. The mounting holes may be oblong or elongated in both directions to provide added positioning flexibility. Components of the control 100 may be configured, e.g., assembled, to be snapped into the lower housing piece or portion 123, and then the top cover or upper housing portion 119 may be attached.

The relay switch control 100 may be provided, e.g., for use in relation to single stage air conditioning and heat pump condensing units with single-phase reciprocating or scroll compressors operating on standard residential and/or commercial (delta and/or wye) power configurations. The control 100 may be used as an aftermarket field upgrade device to replace a traditional contactor, while incorporating additional value-added features, such as short cycle protection, brownout protection, random start delay, cycle count retention and light indicator display. In various exemplary embodiments, a relay switch control is configured to operate using limited indoor unit input, e.g., from only two wires (Y1, C). Additionally, various embodiments may provide for control of a two-stage compressor and thus may include an additional input (Y2) terminal and means for switching a second stage on/off. An example relay switch control may have brownout protection, e.g., similar to that disclosed in U.S. Pat. No. 6,647,346, the entire disclosure of which is incorporated herein by reference.

Various relay switch control embodiments include a single relay for the fan and compressor. But in other exemplary embodiments, a relay switch control may include more than one relay, e.g., as disclosed in U.S. Pat. Nos. 7,100,382, 7,444,824, 7,464,561, and/or 7,694,525, the entire disclosures of which are incorporated herein by reference.

The relay switch control 100 may be used as a field replacement for a standard electromechanical contactor. A typical reason for the failure of standard open frame contactors is the intrusion into the contact area of insects, which foul the contacts and cause the contacts to fail. By using a sealed relay, the insect problem can be avoided and possibly eliminated.

The dipswitches 107 and 109 of the control 100 may be used to provide various features. For example, the first

dipswitch **107** may be used to select/set a short cycle delay of, e.g., 0 or 180 seconds at 60 Hertz, 0 or 216 seconds at 50 Hertz, etc. The second dipswitch **109** may be used to select or deselect brownout protection. In various embodiments, a compressor lockout feature is provided through dipswitch(es). The lockout feature allows an installer to select how many failed attempts to start a compressor connected to the control **100** are to be allowed before the control **100** locks out the compressor. This feature can help protect a compressor and motor from damage, e.g., if a HVAC system needs service. In some embodiments, when a relay switch control locks out the compressor, a message is displayed (e.g., on a thermostat display, etc.) to call for servicing. In some embodiments, a setting for the dipswitch(es) is provided that prevents lock out of the compressor regardless of the number of failed starts.

In the exemplary embodiment of FIG. 3, the tri-color LED indicator **103** may be operable to indicate the number of cycles the relay has cycled, thereby indicating the number of cycles of a compressor and/or other motor connected thereto. Additionally or alternatively, the tri-color LED indicator **103** may be operable to indicate detected faults and/or status conditions. Detected faults may be stored in memory (e.g., within EEPROM of the control, etc.). In various embodiments, a relay switch control may be configured to retain the last stored fault for display if power is lost.

Referring again to the control **100**, faults and status conditions may be indicated, e.g., by color(s) and/or “blink” rate(s) of the LED indicator **103**. Operation of the LED indicator **103** depends on which push button **111** or **113** is depressed. The example control **100** has two push buttons **111**, **113**. The first push button **111** is indicated as “TEST”, and the second push button **113** is indicated as “COUNT”. The operation and function of the tri-color LED indicator **103** may be as shown in the Table shown in FIG. 4 in regard to LED flash codes that may be displayed through the tri-color LED **103**. As shown in FIGS. 3 and 4, the LED flash codes may be slow green flashing for standby, solid green for normal, fast green flashing for delay, green/red flashing for low line voltage, and green/yellow/red flashing for test. Indications of other or additional faults and/or status conditions may be provided in various embodiments.

FIG. 4 also provides terminal designations that may be used for the control **100** (FIG. 3). For example, FIG. 4 shows that for coil details for voltage over 100 VAC, the line level control voltage is connected to “Line”, and the neutral or opposite leg is connected to “Common”. FIG. 8 also shows that for class 2 24 VAC, the 24V control voltage is connected to “Y”, and the 24V ground is connected to “Common”.

By way of example, the operation and functionality of the onboard push buttons **111** and **113** (“TEST” and “COUNT”) shown in FIG. 3 may be as follows.

- A. Pushing the “TEST” button **111** for more than one second provides a five-second powered check of a connected compressor and outdoor fan. Any active delays such as the short cycle timer are bypassed. Any pressure switch lockouts are also cleared by pressing the “TEST” button, and a brownout/low voltage threshold is ignored. The tri-color LED indicator **103** alternates flashing Green-Yellow-Red while the test is active, and then resumes normal displays.
- B. Pushing both the “TEST” and “COUNT” buttons **111**, **113** for more than two seconds activates a fault recall mode. The tri-color LED **103** displays the last fault for ten seconds, after which time the control **100** resumes normal displays. If the “TEST” and “COUNT” buttons

111, **113** are simultaneously pushed for between five and ten seconds, the last fault is cleared, and the control **100** then resumes normal displays.

- C. Pushing the “COUNT” button **113** for more than one second activates a cycle count display mode. In some embodiments, a green LED (i.e., the LED **103** when producing green colored light) represents ten-thousands, a red LED represents thousands, and a yellow LED represents hundreds. Basic numerical rounding may be used to simplify displays. An example of a stored count of 52,518 compressor relay operations would blink out as 5 green, 2 red, 5 yellow. The cycle count is not resettable in the field in this exemplary embodiment. Other or additional push button designations, push button hold times, LED colors, LED blink rates and combinations of the foregoing could be used in various exemplary embodiments.

Referring again to the control **100** shown in FIG. 3, example timing periods may include anti-short-cycle-delay of 0 seconds or 180 seconds (selectable) at 60 Hertz, and 0 seconds or 216 seconds (selectable) at 50 Hertz. Compressor test may be 5 seconds at 60 Hertz and 6 seconds at 50 Hertz. In an exemplary embodiment, the control **100** may be configured to have the following specifications or electrical ratings:

Line voltage input . . . 120/208/240/250 VAC, 50/60 Hz
 Full load amperes . . . (FLA) 40 A
 Locked rotor amperes . . . (LRA) 200 A
 Control (Coil) voltage (Y, C) . . . 24 VAC, 50/60 Hz
 (Line C) . . . 120/208/240/250 VAC, 50/60 Hz

In exemplary embodiments, a relay switch control may include a high-reliability, optically-controlled latching relay, sealed against the intrusion of insects and debris, and that is operable with or across a range of activation inputs, such as activation inputs ranging from 98 VAC to 276 VAC inputs (e.g., 120, 208, 240, 250, 24 VAC inputs, etc.), etc. Various embodiments may provide line voltage brownout protection by de-energizing a compressor, e.g., in the event of calls for compressor operation during line voltage drops. Various embodiments may provide short cycle protection, e.g., by activating a short delay before normal operation for compressors in air conditioners and heat pumps. Relay switch controllers or controls in exemplary embodiments may also detect inputs from high and low pressure switches and lock out compressor operation, e.g., when multiple consecutive pressure switch openings are detected. Additionally or alternatively, example embodiments of relay switch controls may include a cycle counter feature that a user may activate by push button, to determine and display how many times a compressor relay has turned on. Additionally or alternatively, example embodiments of relay switch controls may include a random start delay timer function, e.g., as further described below.

An example embodiment of a relay switch control disclosed herein may be configured for use as a field replacement suitable for replacing any of a plurality of different configurations (e.g., up to 5 ton/40 A, 1-pole, 1.5 pole, 2-pole configurations, etc.) of contactors. Example embodiments may be relatively easy to install, e.g., using lug connectors and a mounting plate that can be installed in the same location previously occupied by a conventional contactor. In various embodiments, relay switch controls may be self-powered and/or may be wired into existing wiring without requiring any new wires.

The control **100** may be self-powered and/or may be configured with a power stealing feature in exemplary embodiments. In various embodiments, the control **100** may

include its own power supply such that an installer is not required to pull additional wires to the outdoor unit.

In addition to being operable across or with a range of activation inputs, such as activation inputs ranging from 98 VAC to 276 VAC inputs (e.g., 120, 208, 240, 250, 24 VAC inputs, etc.), exemplary embodiments may also include or provide one or more of (but not necessarily any or all of) the following features, functions, and benefits. For example, reliability may be improved at least due to one or more of the following features in various embodiments.

- (a) Control relay contacts are enclosed in a seal, thereby preventing insects, debris, and other contaminants from getting into the relay.
- (b) Relay smart “zero cross” switching can inhibit contact damage and improve cycle life.
- (c) Line voltage brownout protection is selectable to deactivate operation in excessively low voltage conditions, on start-up, and/or during run.
- (d) Short cycle protection is selectable, e.g., to maintain equal system pressure conditions.
- (e) The latching relay can reduce or eliminate chatter and can reduce VA draw (e.g., zero chatter latching relay, etc.).

Exemplary embodiments may include a reliable one-million-cycle rated, sealed electronic switch with microprocessor control that inhibits arcing that may otherwise cause contact welding and pitting. For example, FIG. 6 shows an exemplary embodiment of a contactor after one million cycles using microprocessor controlled switching algorithms. By comparison, FIG. 7 shows a traditional contactor after only 300,000 cycles. A comparison of FIGS. 6 and 7 shows the performance improvement (e.g., longer contactor life, over one million cycles, increased reliability, etc.) that may be realized by using microprocessor controlled switching algorithms in accordance with exemplary implementations of the present disclosure. In exemplary embodiments, the switch may be provided in a seal that prevents insects, ants, debris, etc. from entering the switch and saves on pest control treatment.

Exemplary embodiments may allow for one contactor part number to replace the following contactor applications, including (1) AC contactors that operate on 24 VAC that allow for proper operation of legacy thermostats (e.g., mechanical thermostats with anticipation, power stealing); (2) Refrigeration contactors that use 120V, 208V, 240V, or 250V AC coils; and (3) Non-Compressor applications (e.g., motors, fans, etc.) with coils of 24 VAC, 120 VAC, 208 VAC, 240 VAC, or 250 VAC.

Exemplary embodiments may include a door or other covering for the terminals to aid in hook up of a multi-voltage connection. Exemplary embodiments may include a fuse or protective device in the current path of the 24 VAC control signal. The fuse or protective device may protect against miswiring.

Exemplary embodiments may be configured with brownout auto detect implemented via a firmware algorithm (e.g., within the microprocessor 14 (FIG. 1), etc.) for determining input voltage and adjusting brownout threshold.

An example method 221 for determining input voltage and adjusting brownout threshold is disclosed in FIG. 5. As shown in FIG. 5, the method 221 includes, at 223, using complimented EEPROM (broadly, memory) location with 3 separate values to identify 3 conditions. At power up, the control will check EEPROM to determine a brownout threshold value. If the voltage level has not been set (production default value), this will cause the control to determine voltage level at 225. If the voltage level is 120 VAC,

the brownout threshold will be set at 93 VAC for this 120 VAC voltage level. If the voltage level is 240 VAC, the brownout threshold will be set at 180 VAC for this 240 VAC voltage level.

At 225, the method 221 includes determination of voltage range. If the EEPROM locations from 223 show Voltage Level Not Set, the control will monitor input voltage. If the monitored input voltage is determined to be stable within a range from 90 VAC to 140 VAC, then the control will set the EEPROM to the 120 VAC voltage level. And the brownout threshold will be set at 93 VAC for this 120 VAC voltage level. Otherwise, if the monitored input voltage is not determined to be stable within a range from to 140 VAC, then the control will set the EEPROM to the 240 VAC voltage level. And the brownout threshold will be set at 180 VAC for this 240 VAC voltage level.

At 227, the method 221 includes brownout error. If the line voltage remains below the brownout threshold and the dipswitch brownout (e.g., SW2-2 or dipswitch 109 in FIG. 3, etc.) is set to the ON position, the control will display the appropriate error code (e.g., the LED flash code for brownout shown in FIG. 4, etc.). The control will de-energize relay and not allow the relay to resume until the voltage returns to a minimum of 190V for 240V level or 100V for 120V level. If the dipswitch brownout (e.g., SW2-2 or dipswitch 109 in FIG. 3, etc.) is set to the OFF position, the control will not act upon a low VAC line condition.

At 229, the method 221 includes voltage level reset. If the dipswitch brownout (e.g., SW2-2 or dipswitch 109 in FIG. 3, etc.) changes from the OFF state to the ON state, the control will set the voltage level.

Exemplary embodiments of the multi-voltage contactors and controls disclosed herein may be applied to or used with air conditioning systems and/or refrigeration systems. Exemplary embodiments disclosed herein may also be used with or applied to any other application using a contactor to switch an inductive load, such as the inductive loads of electric motors (e.g., pumps, compressors, conveyors, etc.). Although various exemplary embodiments are disclosed herein in relation to a compressor motor, the present disclosure is not so limited. Various embodiments are contemplated in relation to various types of loads, including but not limited to motors that may be switched through contactors/relays, etc. Embodiments of relay switch controls also are contemplated in relation to various types of systems, including but not limited to climate control systems, HVAC, refrigeration systems, pools, spas, hot tubs, water wells, compression based cooling systems, etc. Additionally, although the term “relay switch control” is used herein to refer to various exemplary embodiments, various types of controls, controllers, hardware, software, combinations thereof, etc. could also be used. Various types of processors, microprocessors, computers, etc. could also be utilized in accordance with various implementations of the disclosure.

Example embodiments are provided so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail. In addition, advantages and improvements that may be

achieved with one or more exemplary embodiments of the present disclosure are provided for purpose of illustration only and do not limit the scope of the present disclosure, as exemplary embodiments disclosed herein may provide all or none of the above mentioned advantages and improvements and still fall within the scope of the present disclosure.

Specific dimensions, specific materials, and/or specific shapes disclosed herein are example in nature and do not limit the scope of the present disclosure. The disclosure herein of particular values and particular ranges of values for given parameters are not exclusive of other values and ranges of values that may be useful in one or more of the examples disclosed herein. Moreover, it is envisioned that any two particular values for a specific parameter stated herein may define the endpoints of a range of values that may be suitable for the given parameter (the disclosure of a first value and a second value for a given parameter can be interpreted as disclosing that any value between the first and second values could also be employed for the given parameter). Similarly, it is envisioned that disclosure of two or more ranges of values for a parameter (whether such ranges are nested, overlapping or distinct) subsume all possible combination of ranges for the value that might be claimed using endpoints of the disclosed ranges.

The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. For example, when permissive phrases, such as “may comprise”, “may include”, and the like, are used herein, at least one embodiment comprises or includes the feature(s). As used herein, the singular forms “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

When an element or layer is referred to as being “on,” “engaged to,” “connected to,” or “coupled to” another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

The term “about” when applied to values indicates that the calculation or the measurement allows some slight imprecision in the value (with some approach to exactness in the value; approximately or reasonably close to the value; nearly). If, for some reason, the imprecision provided by “about” is not otherwise understood in the art with this ordinary meaning, then “about” as used herein indicates at least variations that may arise from ordinary methods of measuring or using such parameters. For example, the terms

“generally,” “about,” and “substantially,” may be used herein to mean within manufacturing tolerances. Whether or not modified by the term “about,” the claims include equivalents to the quantities.

Although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.

Spatially relative terms, such as “inner,” “outer,” “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

What is claimed is:

1. A relay switch control configured for connection with one or more components and one or more line voltage sources, the relay switch control comprising:

- a processor;
 - a relay operable by the processor to electrically connect or disconnect the one or more line voltage sources; and
 - a control circuit configured to be coupled for communication with and/or transmit control signals to the processor;
- wherein the processor is configured to be operable for determining input voltage and adjusting brownout threshold; and
- wherein the relay switch control is configured to:
- at power up, determine a voltage level set within a memory;
 - if the voltage level is set at a first voltage level within the memory, set the brownout threshold at a first brownout threshold;
 - if voltage level is set at a second voltage level within the memory, set the brownout threshold at a second brownout threshold; and

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if voltage level is not set within the memory;
 monitor input voltage;
 determine if the monitored input voltage is stable
 within a first voltage range;
 set the voltage level at the first voltage level within
 the memory and set the brownout threshold at the
 first brownout threshold if the monitored input
 voltage is determined to be stable within the first
 voltage range; and
 set the voltage level at the second voltage level
 within the memory and set the brownout threshold
 at the second brownout threshold if the monitored
 input voltage is not determined to be stable within
 the first voltage range.

2. The relay switch control of claim 1, wherein:
 the first voltage level is 120 VAC;
 the first brownout threshold is 93 VAC;
 the second voltage level is 240 VAC;
 the second brownout threshold is 180 VAC; and
 the first voltage range is 90 VAC to 140 VAC.

3. The relay switch control of claim 1, wherein the control
 circuit includes a common terminal, a first terminal, and a
 second terminal, whereby the relay switch control is con-
 figured to be selectively operable with a first voltage input
 and with a second voltage input higher than the first voltage
 input.

4. The relay switch control of claim 3, wherein the relay
 switch control is configured to be selectively operable with:
 a 24 VAC coil when wires of the 24 VAC coil are
 connected to the first terminal and the common termi-
 nal; and
 a line voltage coil when wires of the line voltage coil are
 connected to the second terminal and the common
 terminal.

5. The relay switch control of claim 3, wherein the control
 circuit includes a fuse coupled with and/or in the first
 terminal that is configured to protect against miswiring.

6. The relay switch control of claim 3, wherein the relay
 switch is part of a contactor, and the contactor comprises a
 cover or door movable relative to the first terminal and the
 second terminal, such that only one of the first terminal or
 the second terminal is available for connection at a given
 time.

7. The relay switch control of claim 3, wherein the relay
 switch is part of a contactor, and the contactor comprises a
 cover or door movable relative to the first terminal and the
 second terminal between at least:
 a first position in which the first terminal is exposed and
 available for connection, and the cover or door is

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disposed over the second terminal and inhibits connec-
 tion to the second terminal; and
 a second position in which the second terminal is exposed
 and available for connection, and the cover or door is
 disposed over the first terminal and inhibits connection
 to the first terminal.

8. The relay switch control of claim 1, wherein the control
 circuit includes a common terminal, a first terminal capable
 of receiving a voltage input of 24 VAC, and a second
 terminal capable of selectively receiving a voltage input
 within a range of voltage inputs including 120 VAC, 208
 VAC, 240 VAC, or 250 VAC, whereby the relay switch
 control is configured to be selectively operable with a first 24
 VAC input and with a second 120/208/240/250 VAC input.

9. The relay switch control of claim 1, further comprising
 an AC/DC power supply configured to be operable with
 inputs ranging from 98 VAC to 276 VAC inputs.

10. The relay switch control of claim 1, wherein the relay
 switch is part of a contactor that is configured to be selec-
 tively interchangeable with:
 an AC contactor operable on 24 VAC;
 a refrigeration contactor having 120 VAC, 208 VAC, 240
 VAC, or 250 VAC coils; and
 a non-compressor contactor having 24 VAC, 120 VAC,
 208 VAC, 240 VAC, or 250 VAC coils.

11. The relay switch control of claim 1, wherein the relay
 switch is part of a multi-voltage contactor that is configured
 to accept activation inputs ranging from 98 VAC to 276 VAC
 inputs and/or wherein the multi-voltage contactor is config-
 ured to accept 120, 208, 240, 250, and 24 VAC inputs and
 to switch loads of the same or different voltage.

12. The relay switch control of claim 1, wherein the relay
 switch control is configured for connection with one or more
 components of an HVAC system or other compression based
 cooling system.

13. The relay switch control of claim 1, wherein the relay
 switch control is configured such that:
 if a line voltage remains below the brownout threshold set
 by the relay switch control and a brownout switch of
 the relay switch control is set to the ON position, the
 relay switch control is operable for displaying an
 indication of an error and de-energizing the relay of the
 relay switch control to inhibit operation until a line
 voltage returns to a predetermined minimum level; or
 if the brownout switch of the relay switch control is set to
 the OFF position, the relay switch control will not act
 upon a low voltage line condition.

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